

From Cosine to WSD: Identifying Transferable LR-Drop Response in MPL Residuals

Source-only calibration without target WSD loss

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Peking University Deep Learning Course Task 2
github.com/wjjpku/DL-final

Central question

Given a cosine training loss curve and a target WSD-family schedule, can we predict the full WSD loss curve without reading the target WSD loss?

Thesis

Cosine residuals contain both a **transferable LR-drop response** and a **non-transferable MPL-LD parameter error**. The contribution is to identify these two components and transfer only the former.

How to read this report

This report does not start from “adding a kappa correction”. It starts from a residual identification problem:

MPL residual transfer is hard because residuals are confounded.

Link	Question	Evidence
Baseline	Are the baseline laws reproducible?	Tissue/Momentum and MPL reproduction.
Obstacle	Can cosine residual be transferred directly?	Direct-kappa negative control.
Identification	How do we remove nuisance drift?	MPL-LD projection.
Prediction	How do we build a target-free correction?	Schedule-only ϕ plus source-only $\hat{\kappa}$.
Validation	Is the identified component transferable?	15/15 WSD-family wins and oracle-kappa alignment.

Reading order. First understand why residuals are confounded, then how projection identifies the transferable part, and finally how each experiment validates one link in this chain.

Problem setting: source cosine $\rightarrow$ target WSD

Inputs.

- source: a cosine learning-rate schedule and its observed training loss curve;
- baseline: frozen MPL predictions for both source and target schedules;
- target: a WSD-family learning-rate schedule $\{\eta_t\}_{t=1}^T$.

Output.

$$\widehat{L}_s(t), \quad t = 1, \dots, T.$$

Restriction.

calibration uses cosine residual only, target loss is used only for evaluation.

Why this is not ordinary curve fitting. Cosine and WSD schedules have different decay geometry. If a source residual component transfers to WSD, it must correspond to a schedule-response mechanism, not just a low-frequency fitting error on the source curve.

Data split and experiment setting

Model scales.

25M, 100M, 400M.

Source curve.

$$\text{cosine_72000.csv}, \quad r_{\cos}(t) = L_{\cos}(t) - L_{\text{MPL},\cos}(t).$$

The residual calibration uses the source suffix $t \geq 8000$; the window is chosen by a source-side identifiability check.

WSD-family targets.

File	Meaning
wsd_20000_24000.csv	WSD sharp cooldown
wsdld_20000_24000.csv	WSD linear decay
wsdcon_3.csv	WSD-con final LR 3×10^{-5}
wsdcon_9.csv	WSD-con final LR 9×10^{-5}
wsdcon_18.csv	WSD-con final LR 18×10^{-5}

Main evaluation. Same-scale WSD-family transfer: 3 scales $\times$ 5 targets, giving 15 target loss curves.

Data processing pipeline

- 1 Read source cosine loss and LR schedule.
- 2 Read frozen MPL prediction.
- 3 Compute source residual:

$$r_{\cos}(t) = L_{\cos}(t) - L_{\text{MPL},\cos}(t).$$

- 4 Use calibration suffix $t \geq 8000$.
- 5 For each target WSD schedule, compute LR drops d_t and response feature ϕ .
- 6 Construct MPL-LD tangent basis and projection P_{LD} .
- 7 Estimate source-only $\hat{\kappa}$.
- 8 Predict target loss; target loss is used only after prediction for evaluation.

Fixed protocol

- source suffix: $t \geq 8000$
- logging interval: $\Delta_{\text{obs}} = 128$
- response time: q_2 half-life rule
- ridge floor: $1/N_{\text{cal}}$
- constraint: $\kappa \geq 0$
- metrics: MAE, RMSE, terminal error

Data source. All main loss curves are stored under `external/MultiPowerLaw/...`, with subfolders `csv_25`, `csv_100`, and `csv_400`.

Strong baseline: MPL is frozen and reproducible

The frozen MPL baseline is:

$$L_{\text{MPL},s}(t) = L_{0,s} + A_s S_s(t)^{-\alpha_s} + B_s D_s(t; C_s, \beta_s, \gamma_s).$$

This work studies the MPL residual:

$$r_s(t) = L_s(t) - L_{\text{MPL},s}(t).$$

Scale	R2	MAE	RMSE	PredE	WorstE
25M	0.998802	0.003760	0.004652	0.001102	0.004095
100M	0.998301	0.004348	0.005919	0.001425	0.005829
400M	0.997762	0.004835	0.007305	0.001679	0.009948

Role. The reproduction metrics are consistent with the MPL README. Therefore the correction is studied on top of a strong MPL baseline, not on top of a weak baseline.

Reproduction details

The reproduced object is the public MPL experiment / frozen MPL prediction. This work does not retrain MPL and does not refit MPL on WSD target loss. The main difficulties are logging-grid alignment, scale-file matching, and finite-difference construction of the MPL-LD tangent basis. The key difference is that after reproducing MPL, this work only corrects its residual using source-only calibration.

Baseline reproduction: Tissue et al. / Momentum Law

In addition to MPL, I reproduced the learning-rate annealing baseline from Tissue et al. / Momentum Law. In the repository this baseline is implemented as `tissue` in `repro/reproduce_cosine_to_wsd.py`.

$$L_{\text{Tissue}}(t) = L_0 + AS_1(t)^{-\alpha} - CS_2(t), \quad S_2(t) = \sum_{\tau \leq t} m_\tau, \quad m_\tau = \lambda m_{\tau-1} + (\eta_{\tau-1} - \eta_\tau).$$

What is being reproduced

This is the one-pole annealing / momentum response baseline: LR drops are accumulated through a first-order state m_t , then added as the schedule-dependent term $S_2(t)$. It is a low-capacity baseline rather than a residual-identification method.

Fixed reproduction protocol

Fit on source cosine curves only: `cosine_24000.csv` and `cosine_72000.csv`; then evaluate on the same 15 WSD-family targets used in this report. No WSD target loss is used during fitting.

Command. `python3 repro/reproduce_cosine_to_wsd.py`

Reproduction result: Momentum Law is a valid but weaker WSD baseline

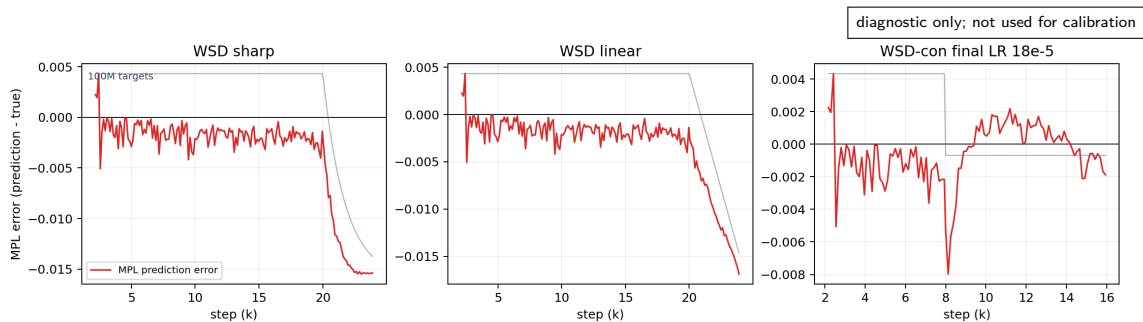
The just-reproduced Tissue/Momentum result is suitable for the baseline section: it verifies an earlier LR-annealing law under the same source-only protocol, but it does not replace MPL as the backbone for residual identification.

Baseline	Cosine train MAE	WSD test MAE	WSD test RMSE	WSD test R^2
Tissue et al. / Momentum Law	0.002242	0.007493	0.010415	0.995223
MPL	0.003517	0.006292	0.009848	0.995760

Scale	Tissue/Momentum			MPL		
	MAE	RMSE	R^2	MAE	RMSE	R^2
25M	0.007831	0.009842	0.995576	0.005388	0.008573	0.996521
100M	0.004859	0.007490	0.997855	0.004964	0.008152	0.997130
400M	0.009790	0.013913	0.992238	0.008524	0.012820	0.993629

Interpretation. Tissue/Momentum fits the source cosine curves more tightly ($0.002242 < 0.003517$), but MPL gives the stronger average WSD-family transfer baseline ($0.006292 < 0.007493$). Therefore the residual-identification method is built on frozen MPL, while Tissue/Momentum is retained as a reproduced baseline reference.

Diagnostic observation: MPL errors concentrate near transition / tail



Reading the figure. The red curve is MPL prediction error and the gray curve is normalized LR. This page reads target true loss only to show the error shape; it is not used for $\hat{\kappa}$ calibration. For WSD sharp / linear, MPL error concentrates after the LR decay. For WSD-con, different final LR's induce different residual shapes.

Question. These errors look schedule-related, but does the source cosine residual contain the same transferable component?

Key obstacle: cosine residual is a mixed signal

If we directly fit a response amplitude from the cosine residual,

$$\hat{\kappa}_{\text{direct}} = \frac{\langle \phi_{\lambda, \cos}, r_{\cos} \rangle_+}{\|\phi_{\lambda, \cos}\|_2^2 + 1/N_{\text{cal}}},$$

we implicitly assume:

$$r_{\cos}(t) \approx \kappa \phi_{\lambda, \cos}(t).$$

This assumption is too strong.

A more realistic local decomposition is:

$$r_{\cos}(t) = \underbrace{\kappa \phi_{\lambda, \cos}(t)}_{\text{transferable LR-drop response}} + \underbrace{J_{\text{LD}}(t) \Delta \theta_{\text{LD}}}_{\text{non-transferable MPL-LD nuisance}} + \epsilon(t)$$

Intuition. Cosine decay is smooth, so $\phi_{\lambda, \cos}$ is low-frequency. MPL-LD parameter error also creates low-frequency drift. Without separating them, κ absorbs non-transferable MPL parameter error.

Negative control: direct cosine κ fails

Estimator	Pearson vs κ^*	kappa RMSE	mean MAE change	wins
direct no-projection kappa	-0.849	0.6294	+625.92%	0/15
projected kappa	+0.910	0.0103	-30.88%	15/15

Conclusion. The core problem is not whether we can add a residual correction to MPL. The core problem is **whether we can identify the transferable part inside the source residual**.

Task reformulation

Separate the following two components in MPL residuals:

transferable LR-drop response vs. MPL-LD nuisance drift.

Transfer only the former.

Core idea: project out MPL-LD tangent nuisance

The MPL learning-rate dependent term depends on:

$$\theta_{\text{LD}} = (B, C, \beta, \gamma).$$

If these parameters have small errors, the first-order MPL prediction error is:

$$\Delta L_{\text{MPL}}(t) \approx J_{\text{LD}}(t) \Delta \theta_{\text{LD}},$$

where:

$$J_{\text{LD}}(t) = \left[\frac{\partial L_{\text{MPL}}}{\partial \log B}, \frac{\partial L_{\text{MPL}}}{\partial \log C}, \frac{\partial L_{\text{MPL}}}{\partial \log \beta}, \frac{\partial L_{\text{MPL}}}{\partial \log \gamma} \right].$$

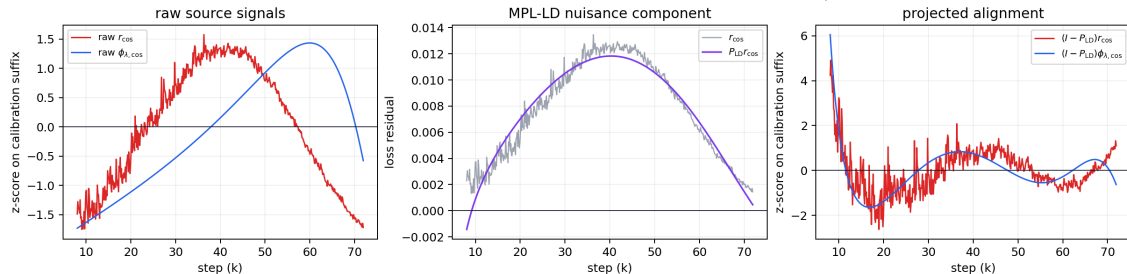
Let P_{LD} be the orthogonal projection onto the span of J_{LD} . Before estimating response amplitude, use:

$$x = (I - P_{\text{LD}})\phi_{\lambda, \cos}, \quad y = (I - P_{\text{LD}})r_{\cos}.$$

Meaning. Estimate the transferable response only in residual directions that local MPL-LD parameter perturbations cannot explain.

What projection removes from cosine residual

100M source cosine, calibration suffix $t \geq 8000$; lambda from WSD sharp



Reading the figure. The left panel shows that raw cosine residual and raw response feature are both low-frequency. The middle panel shows that the MPL-LD tangent space explains much of the smooth residual drift. The right panel shows that after nuisance removal, the estimated κ comes from the alignment between projected residual and projected feature.

Conclusion. Projection is not a decorative linear-algebra step. It removes non-transferable MPL-LD drift from the source cosine residual.

Prediction rule: minimal deployable model

After projected identification, WSD prediction uses:

$$\widehat{L}_s(t) = L_{\text{MPL},s}(t) + \widehat{\kappa}_s \phi_{\lambda_s,s}(t)$$

Part	Meaning
$L_{\text{MPL},s}(t)$	frozen MPL baseline.
$\phi_{\lambda_s,s}(t)$	causal LR-drop response shape induced by the target schedule.
λ_s	response decay rate fixed by the target schedule using the q_2 half-life rule.
$\widehat{\kappa}_s$	the only scalar estimated from loss residual; it uses source cosine residual only.

One sentence. The target schedule determines the response shape; the source residual determines the response amplitude. WSD prediction does not read WSD loss.

Causal LR-drop response: schedule-only shape

Define positive LR drop:

$$d_t = [\eta_{t-1} - \eta_t]_+.$$

When LR does not decrease, $d_t = 0$.

Define the response recursion:

$$z_t = \exp(-\lambda_s \eta_t) z_{t-1} + d_t, \quad \phi_{\lambda_s, s}(t) = \frac{z_t}{\eta_{\max}}.$$

Equivalently:

$$\phi_{\lambda_s, s}(t) = \frac{1}{\eta_{\max}} \sum_{u \leq t} d_u \exp \left(-\lambda_s \sum_{v=u+1}^t \eta_v \right).$$

Interpretation. An LR drop is a forcing event. Each drop injects a residual impulse, and the impulse decays in cumulative LR time. Therefore ϕ is schedule-only, causal, and target-loss-free.

Response time scale. λ_s is fixed by a schedule-only half-life rule: diffuse LR decay receives a longer response time, while a sharp LR drop receives a shorter response time.

On the source cosine suffix, define:

$$x_s = (I - P_{LD})\phi_{\lambda_s, \cos}, \quad y = (I - P_{LD})r_{\cos}.$$

The estimation problem is:

$$\min_{\kappa \geq 0} \|y - \kappa x_s\|_2^2 + \frac{1}{N_{\text{cal}}} \kappa^2.$$

Closed-form solution:

$$\hat{\kappa}_s = \frac{[x_s^\top y]_+}{\|x_s\|_2^2 + 1/N_{\text{cal}}}$$

Interpretation. $\hat{\kappa}_s$ is the amplitude of the projected response feature. It is not a multi-parameter residual regressor. If x_s and y have no positive alignment after projection, the response is not activated.

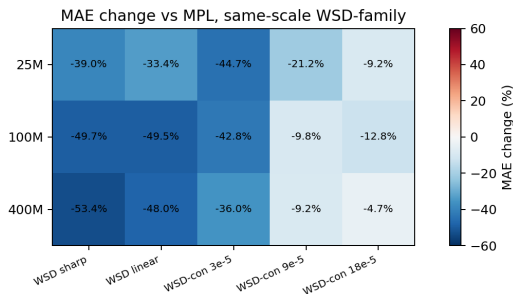
Target-leakage audit: which quantities read target loss

source cosine residual $\rightarrow$ projected $\hat{\kappa}$; target LR schedule $\rightarrow \phi$; frozen MPL $+ \hat{\kappa}\phi \rightarrow$ prediction; target loss $\rightarrow$ evaluation only.

Stage	Used quantities	WSD loss?	Role
Calibration	<code>cosine_72000.csv</code> , cosine LR, MPL cosine prediction	no	compute source residual and $\hat{\kappa}$
Target feature	target LR schedule	no	compute q_2 , λ_s , $\phi_s(t)$
Prediction	target MPL prediction, $\phi_s(t)$, source-only $\hat{\kappa}$	no	output $L_{\text{MPL}} + \hat{\kappa}\phi$
Evaluation	target loss curve	yes	compute MAE and terminal error
Oracle diagnostics	target residual / target MAE grid	yes	only for κ^* , oracle λ

Conclusion. The deployable prediction stage does not read target WSD loss. Any target-loss-dependent quantity is explicitly marked as evaluation or oracle diagnostic.

Claim 1: target-free correction improves WSD prediction

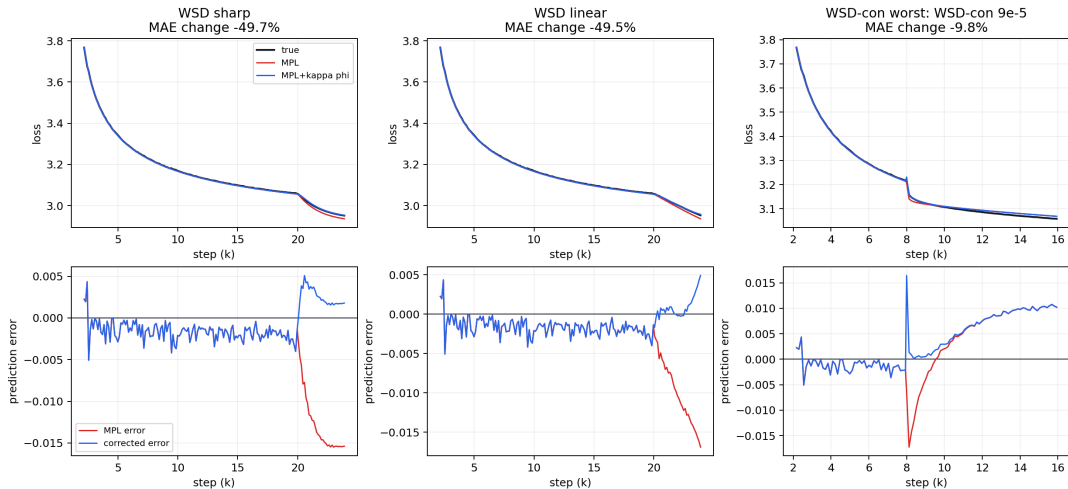


Mean MAE change	Worst row	Wins	Target-loss calibration
-30.88%	-4.67%	15/15	0

Reading the figure. Each cell is the MAE change relative to MPL for one target loss curve. Blue indicates lower error.

Meaning. Without reading target WSD loss, the identified LR-drop response reduces MAE on all 15 WSD-family curves.

Representative curves: correction acts near transition / tail



Reading the figure. The top row compares true loss, MPL, and $L_{\text{MPL}} + \widehat{\kappa}\phi$. The bottom row shows prediction error. The correction mainly acts near the LR transition and tail, instead of arbitrarily shifting the whole curve.

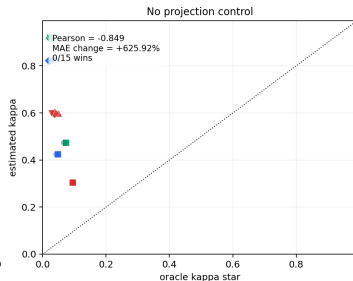
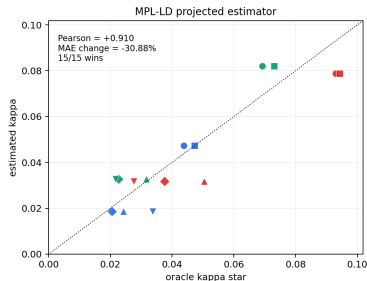
Claim 2: source $\hat{\kappa}$ aligns with target oracle κ^*

Oracle diagnostic.

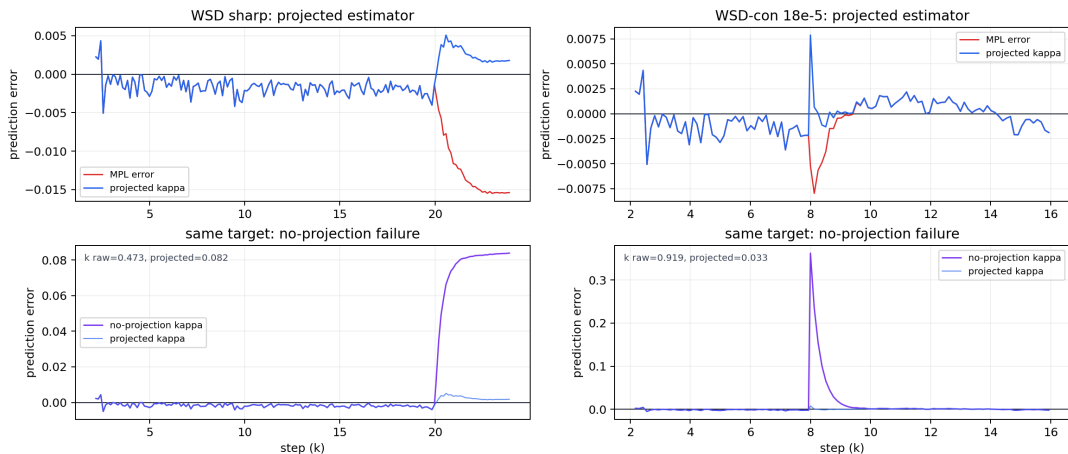
$$\kappa_s^* = \frac{[\langle \phi_{\lambda_{s,s}}, L_s - L_{\text{MPL},s} \rangle]_+}{\|\phi_{\lambda_{s,s}}\|_2^2}.$$

κ^* reads target loss, so it cannot be used for prediction. It only checks whether source-only $\hat{\kappa}$ is close to the target-own optimum.

Result. After projection, Pearson is +0.910; the no-projection control is -0.849.



Projection ablation: with projection / without projection



Reading the figure. The top row compares MPL error with projected $\hat{\kappa}$. The bottom row shows that no-projection κ treats low-frequency drift in the source residual as response amplitude, causing large target error shifts.

Conclusion. Projection is not merely improving a table score. It prevents non-transferable MPL-LD nuisance from being transferred to WSD.

Cross-scale transfer: amplitude has signal, raw residuals do not merge safely

Protocol	Pearson	mean MAE change	worst	wins
same scale	+0.910	-30.88%	-4.67%	15/15
leave-one-scale-out mean κ	+0.671	-25.62%	-3.96%	15/15
all-scale mean κ	+0.808	-29.15%	-4.20%	15/15
single external source scale	+0.731 ~ +0.921	-21.90% ~ -27.06%	-3.15%	10/10 each
pooled raw residuals	+0.641	+62.57%	+231.45%	1/15

Interpretation. Across scales, estimating κ per scale and then aggregating is more reliable than directly concatenating residuals. The response amplitude has cross-scale signal, but absolute residuals should not be pooled naively.

Audit: why this is not high-capacity residual fitting

The residual model is deliberately low-capacity:

$$\widehat{L}_s(t) = L_{\text{MPL},s}(t) + \widehat{\kappa}_s \phi_{\lambda_s,s}(t).$$

Module	Source	residual-fitted parameters
$\phi_{\lambda_s,s}$	target LR schedule	0
λ_s	schedule-only half-life rule	0
P_{LD}	MPL tangent basis	0
$\widehat{\kappa}_s$	source cosine residual projection	1

Risk control. There is no target-family channel selection, no arbitrary basis expansion, and no target-loss-tuned residual regressor. The main overfitting risks are the calibration window and schedule-family coverage.

Failure mode: WSD-con final-LR ranking remains weak

Issue	Observation
What works	All 9/9 WSD-con curves improve MAE, so the main step/tail response is captured.
What remains weak	Within WSD-con, fine-grained final-LR ranking is weak: $\hat{\kappa}$ does not fully distinguish amplitudes across $3e-5$, $9e-5$, $18e-5$.
Next evidence needed	More WSD-con final-LR runs, or a scale/final-LR constrained amplitude model.

Interpretation. The current ϕ captures the main transition/tail response, but not all final-LR-dependent amplitude variation. The complete numeric table is in the appendix.

What this work can claim

- On top of a frozen MPL baseline, source cosine residual contains transferable LR-drop response amplitude.
- MPL-LD tangent projection is the key identification step.
- Same-scale WSD-family prediction improves on all 15 curves.
- $\hat{\kappa}$ aligns with target oracle κ^* with Pearson +0.910.
- Leave-one-scale-out mean- κ cross-scale transfer still gives 15/15 wins.

What this work should not claim

- It is not a universal training-loss law.
- $t \geq 8000$ is not claimed to be a universal constant.
- WSD-con final-LR fine ranking is not fully solved.
- The method is not yet validated on all LR schedule families.

Remaining validation gap: new held-out WSD runs

Current evidence limit

For each scale, the repository contains five WSD-family targets: sharp, linear, WSD-con $3e-5$, WSD-con $9e-5$, and WSD-con $18e-5$. Existing ablations support projection, λ , kernel, and cross-scale behavior, but they cannot replace genuinely new held-out schedules.

Most important next step

The remaining validation gap is not another ablation on existing curves, but new held-out WSD-family training runs after freezing the protocol.

Frozen rules.

$$t \geq 8000, \quad q_2 \text{ half-life}, \quad P_{LD}, \quad 1/N_{\text{cal}} \text{ ridge floor}, \quad \kappa \geq 0.$$

New schedule dimensions. cooldown start, decay length, final LR, and intermediate shapes between sharp and linear.

- ① **Problem formulation.** Cosine-to-WSD residual transfer is formulated as an identification problem on top of frozen MPL:
transferable LR-drop response vs. MPL-LD nuisance drift.
- ② **Identification method.** MPL-LD tangent projection removes local parameter-error directions in the source residual, and a one-dimensional nonnegative ridge projection gives deployable $\hat{\kappa}$.
- ③ **Evidence.** Projected source $\hat{\kappa}$ aligns with target oracle κ^* and reduces MAE on 15/15 same-scale WSD-family curves, while the no-projection negative control fails systematically.

The core contribution is residual identification, not feature stacking.

Assignment checklist

Requirement	Location
1. Background and goal: research content, motivation, and core question.	slides 1–3
2. Data processing and setup: source data, pipeline, model/method, and parameters.	slides 4–8, 14–16
3. Reproduction result: reproduced baseline and key differences/difficulties.	slides 6–8
4. Method and result comparison: proposed method, results, and baseline comparison.	slides 10–22
5. Analysis and discussion: results, failure mode, scope, and validation gap.	slides 23–26
6. Code and division: repository, scripts, environment, outputs, and work division.	last slide

Note. The main text follows the residual-identification story. The appendix keeps q_2 details, kernel ablation, calibration window, MPL L_G audit, and the full failure-mode table.

The key to cosine-to-WSD transfer is not fitting residuals harder, but identifying which residual component is transferable LR-drop response.

$$\hat{L}_s(t) = L_{\text{MPL},s}(t) + \hat{\kappa}_s \phi_{\lambda_s,s}(t)$$

Main result

$\hat{\kappa}$ vs. κ^* : Pearson = +0.910.

same-scale WSD-family: − 30.88% mean MAE change, 15/15 wins.

Appendix A: q2 half-life rule

Loss is observed only on discrete logging points. Let the main logging interval be:

$$\Delta_{\text{obs}} = 128, \quad \lambda_{\text{obs}} = \frac{\log 2}{\eta_{\text{max}} \Delta_{\text{obs}}}.$$

Let total drop and normalized drop mass be:

$$D_s = \sum_t d_t, \quad p_t = \frac{d_t}{D_s}.$$

Define Herfindahl concentration:

$$q_s = \sum_t p_t^2 = \frac{1}{n_{\text{eff}}}.$$

This work uses:

$$\lambda_s = \frac{\lambda_{\text{obs}}}{2 - q_s}$$

Intuition. Diffuse decay receives roughly a $2\Delta_{\text{obs}}$ half-life; a single sharp drop receives roughly a Δ_{obs} half-life. The rule reads only the schedule and not target loss.

Appendix B: Why one-pole exponential response?

This work treats LR drop as forcing and residual relaxation as a low-capacity one-pole closure:

$$z_t = \exp(-\lambda_s \eta_t) z_{t-1} + d_t.$$

Three points:

- ① LR drop is a discrete forcing: $d_t = [\eta_{t-1} - \eta_t]_+$.
- ② If recovery from forcing is approximated by one state variable, the state decays in cumulative LR time.
- ③ The Green's function of a first-order linear system is an exponential kernel:

$$\exp \left(-\lambda_s \sum_{v=u+1}^t \eta_v \right).$$

Boundary. This is not a strict training-dynamics theorem. It is a one-pole closure assumption. Kernel ablations test whether it is more reasonable than same-capacity alternatives.

λ sensitivity	Rule	mean MAE	wins
	q_2 half-life	-30.88%	15/15
	fixed $1\Delta_{\text{obs}}$	-17.52%	15/15
	fixed $2\Delta_{\text{obs}}$	+20.63%	7/15
	wrong fast $\lambda = 20$	-14.76%	15/15
	oracle grid best	-31.55%	15/15

Same-capacity kernel	One-scalar feature	mean MAE	wins
	LR-time exp drop	-30.88%	15/15
	signed LR-time	-30.24%	15/15
	step-time exp	-3.05%	15/15
	cumulative drop	+27.15%	7/15
	power-law LR-time	+119.07%	1/15

Reading the table. Oracle grid best reads target loss and is only an upper-bound diagnostic. The q_2 half-life rule does not read target loss but is close to the oracle. The kernel audit shows that not every schedule-derived scalar explains residual transfer.

Appendix D: source-only calibration window rule

Candidate suffixes are checked using source cosine only:

$$\rho = \frac{\|(I - P_{LD})\phi\|_2^2}{\|\phi\|_2^2}, \quad \rho \leq 1/N_{\text{cal}}.$$

fit start	max retention	floor $1/N_{\text{cal}}$	pass	target result
5000	0.013748	0.001912	no	+139.64%, 0/15
6500	0.004225	0.001953	no	+21.07%, 4/15
8000	0.001478	0.002000	yes	-30.88%, 15/15
10000	0.000489	0.002066	yes	-9.85%, 15/15
12000	0.000353	0.002132	yes	-1.82%, 6/15

Reading the table. The rule selects the earliest passing suffix, $t \geq 8000$. Target loss is not used to choose the window; the sweep is reported only as a post-hoc sensitivity analysis.

Appendix E: why not just rescale the last MPL term?

The last MPL term $L_G(t) = B_s D_s(t)$ is also schedule-aware. A natural suspicion is:

$$\text{proposed correction} \approx \rho L_G(t).$$

Scalar multiplier audit:

Setting	proposed correction	oracle scalar ρL_G	scalar R^2
all core	about -30%	-8.91%	$+0.039$
WSD final	about -45%	-17.80%	$+0.036$
WSD-con	-21.14%	-2.98%	$+0.040$

Conclusion. Rescaling the last MPL term is a meaningful ablation, but it does not explain the main correction shape. ϕ is a causal LR-drop response, not a constant multiple of $L_G(t)$.

Appendix F: WSD-con failure-mode numbers

scale	target	$\hat{\kappa}$	κ^*	ratio	MAE change
25M	3e-5	0.0186	0.0243	0.766	−44.65%
25M	9e-5	0.0186	0.0206	0.904	−21.19%
25M	18e-5	0.0186	0.0338	0.551	−9.19%
100M	3e-5	0.0327	0.0317	1.032	−42.76%
100M	9e-5	0.0327	0.0227	1.444	−9.84%
100M	18e-5	0.0327	0.0217	1.506	−12.80%
400M	3e-5	0.0318	0.0504	0.631	−35.97%
400M	9e-5	0.0318	0.0376	0.845	−9.17%
400M	18e-5	0.0318	0.0277	1.149	−4.67%

Reading the table. Every WSD-con curve improves aggregate MAE, but the $\hat{\kappa}/\kappa^*$ ratio is not stable across final LR.

Appendix G: missing new held-out WSD schedules

For each scale, the currently available WSD-family targets are:

{sharp, linear, WSD-con 3e-5, WSD-con 9e-5, WSD-con 18e-5}.

Therefore the current held-out / robustness analysis can only audit the protocol within these curves.

Most important next experiment. After freezing the following rules, add 2–4 new WSD-family training runs:

$$t \geq 8000, \quad q_2 \text{ half-life}, \quad P_{\text{LD}}, \quad 1/N_{\text{cal}} \text{ ridge floor}, \quad \kappa \geq 0.$$

The new schedules can vary:

- cooldown start;
- decay length;
- final LR;
- intermediate shapes between sharp and linear.

Purpose. Test whether the identification protocol was tuned to the existing five WSD target types.

Artifact	Content
<code>robustness_audit.py</code>	λ sensitivity, kernel ablation, cross-scale, window, failure mode, and figures.
<code>REPORT.md</code>	Complete tables and interpretation for the robustness audit.
<code>LEAKAGE_AUDIT.md</code>	Target-leakage checklist.
<code>slides/figs/</code>	Heatmap, representative error curves, MPL residual anomaly, projection decomposition, projection ablation, and kappa scatter.

Reproduction command.

```
python3 repro/schedule_response_robustness_audit.py  
cd slides && pdflatex main.tex && pdflatex main.tex
```

GitHub:

github.com/wjjpku/DL-final

Main scripts

- reproduce MPL baseline:
`python3 repro/reproduce_cosine_to_wsd.py`
- run MPL-LD projected kappa:
`python3
repro/schedule_response_robustness_audit.py`
- generate figures / tables:
`python3
repro/schedule_response_robustness_audit.py`

Required environment

- Python 3
- numpy, pandas, scipy, matplotlib
- scikit-learn, tqdm, requests
- pdfLaTeX or XeLaTeX for slides

Outputs

- MPL reproduction metrics
- WSD-family MAE / RMSE tables
- kappa-oracle scatter plots
- projection ablation figures
- slides and audit reports

Division

Individual completion

This project was completed individually by Jiaju Wu: method design, code implementation, experiment execution, figure/table generation, and report writing. The MPL baseline reproduction is based on public implementation / public data; citations and modifications are documented in the repository.